

CS & IT ENGINEERING

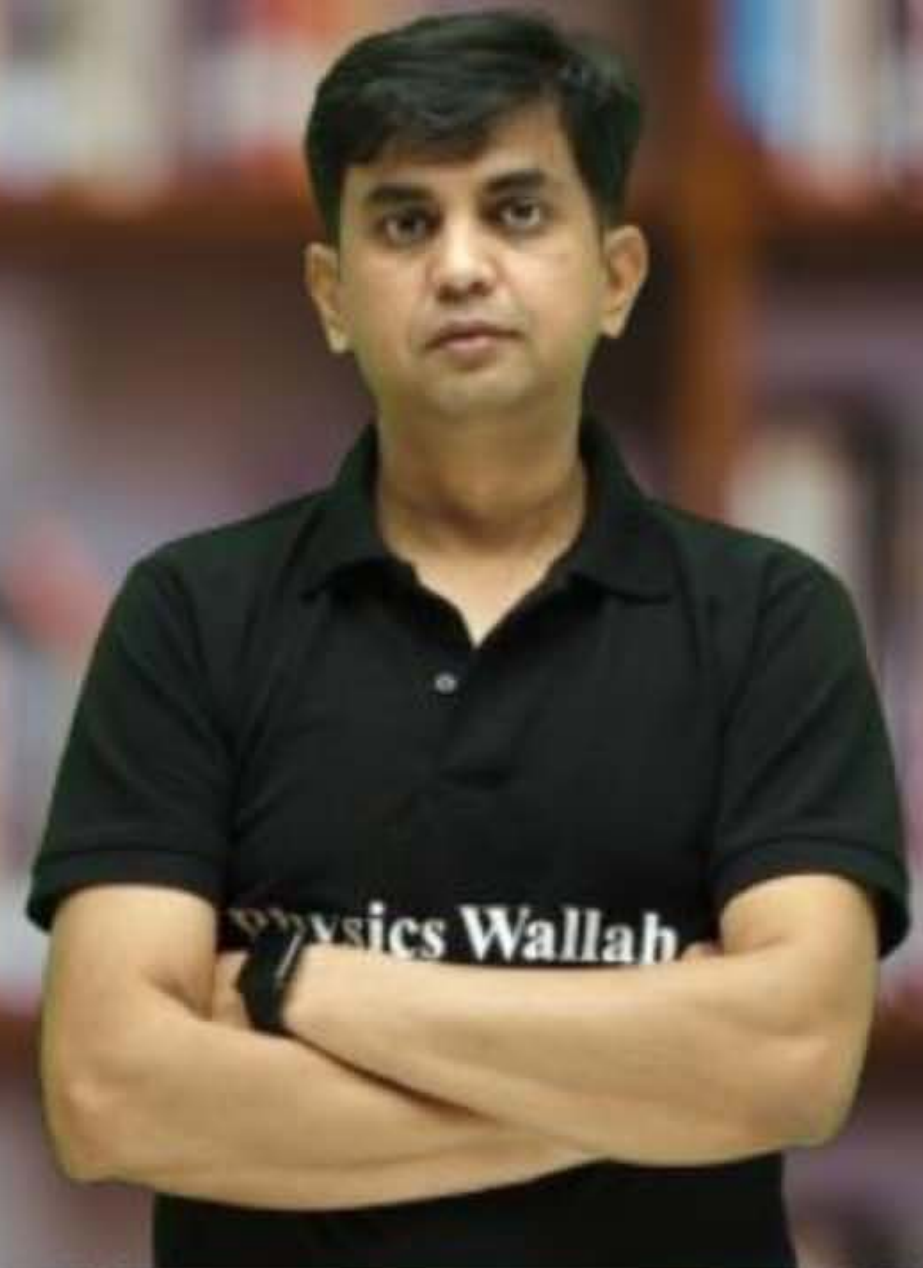


Computer Network

IPv4 Address

Lecture No. – 05

By - Abhishek Sir





Recap of Previous Lecture



Topic

Subnetting





Topics to be Covered



Topic

VLSM

Topic

CIDR

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

Telegram Link : https://t.me/abhisheksirCS_PW



[MCQ]

[GATE-2019][2 Mark]



#Q. Consider three machines M, N and P with IP addresses "100 . 10 . 5 . 2", "100 . 10 . 5 . 5" and "100 . 10 . 5 . 6" respectively. The subnet mask is set to "255 . 255 . 252" for all the three machines. Which one of the following is true?

- ☒ A M, N and P all belong to the same subnet
- ☐ B Only N and P belong to the same subnet
- ☒ C Only M and N belong to the same subnet
- ☒ D M, N, and P belong to three different subnets

$$252 \rightarrow \underbrace{11111100}$$

2	$\rightarrow$	<u>00000010</u>	(M)
5	$\rightarrow$	<u>00000101</u>	(N)
6	$\rightarrow$	<u>00000110</u>	(P)

Ans: B

Subnet Mask :

$$255 . 255 . 255 . 252 = 255 . 255 . 255 . 11111100$$

$$M : 100 . 10 . 5 . 2 = 100 . 10 . 5 . 00000010$$

$$N : 100 . 10 . 5 . 5 = 100 . 10 . 5 . 00000101$$

$$P : 100 . 10 . 5 . 6 = 100 . 10 . 5 . 00000110$$

[MCQ]

[GATE-IT-2006] [2 Mark]



#Q. A subnetted Class B network has the following broadcast address : 144 . 16 . 95 . 255. Its subnet mask

- ☒ A is necessarily 255 . 255 . 224 . 0
- ☒ B is necessarily 255 . 255 . 240 . 0
- ☒ C is necessarily 255 . 255 . 248 . 0
- ☒ D could be any one of 255 . 255 . 224 . 0, 255 . 255 . 240 . 0, 255 . 255 . 248 . 0

Ans: D

Class B Network :-

Subnet Directed Broadcast Address :

$$144.16.95.255 = 144.16.01011111.11111111$$

NetID 16 bits Min^m 3 bit subnetID Max^m 13 bit HostID

3-bit subnetting

$$255.255.224.0 = 255.255.11100000.00000000$$

4-bit subnetting

$$255.255.240.0 = 255.255.11110000.00000000$$

$$255.255.248.0$$

5-bit subnetting

$$255.255.11111000.00000000$$

Example 15 :-

Classless network (Network Address : 40 . 50 . 64 . 0 / 19), with 4-bit subnetting.

Network Address : 40 . 50 . 64 . 0 / 19

Broadcast Address : 40 . 50 . 95 . 255 / 19
(Network Directed)

01000000.00000000
13 bit Host ID
01011111.11111111

Before Subnetting :-

First Host IP Address : 40 . 50 . 64 . 1 / 19

Last Host IP Address : 40 . 50 . 95 . 254 / 19

Network Size = $(2^{13} - 2)$ hosts

After 4-bit Subnetting :-

Sub-network Address

19 bit NetID 4 bit subnetID 9 bit HostID

: 40.50.010 _ _ _ _ 0.00000000 / 23

First Host IP Address

: 40.50.010 _ _ _ _ 0.00000001 / 23

Last Host IP Address

: 40.50.010 _ _ _ _ 1.11111110 / 23

Subnet Broadcast Address
(Sub-network Directed)

: 40.50.010 _ _ _ _ 1.11111111 / 23

First Sub-network Address : 40.50.01000000.00000000 / 23

: 40.50.64.0 / 23

First Host IP Address : 40.50.01000000.000000001 / 23

: 40.50.64.1 / 23

Last Host IP Address : 40.50.01000001.11111110 / 23

: 40.50.65.254 / 23

First Subnet Broadcast Address : 40.50.01000001.11111111 / 23

(Sub-network Directed) : 40.50.65.255 / 23

1st subnet $\Rightarrow \Rightarrow 40.50.64.0$



64.255

65.0

65.1

65.254

$\Rightarrow 40.50.65.255$

2^8

2^8

2^9 Address

Second Sub-network Address : 40.50.01000010.00000000 / 23

: 40.50.66.0 / 23

First Host IP Address : 40.50.01000010.000000001 / 23

: 40.50.66.1 / 23

Last Host IP Address : 40.50.01000011.11111110 / 23

: 40.50.67.254 / 23

Second Subnet Broadcast Address: 40.50.01000011.11111111 / 23

(Sub-network Directed) : 40.50.67.255 / 23

Last Sub-network Address : 40.50.01011110.00000000 / 23

: 40.50.94.0 / 23

First Host IP Address : 40.50.01011110.000000001 / 23

: 40.50.94.1 / 23

Last Host IP Address : 40.50.01011111.11111110 / 23

: 40.50.95.254 / 23

Last Subnet Broadcast Address : 40.50.01011111.11111111 / 23

(Sub-network Directed) : 40.50.95.255 / 23

After 4-bit Subnetting :-

First Sub-network Address : 40 . 50 . 64 . 0 / 23

Second Sub-network Address : 40 . 50 . 66 . 0 / 23

Third Sub-network Address : 40 . 50 . 68 . 0 / 23

Fourth Sub-network Address : 40 . 50 . 70 . 0 / 23

⋮

15th Sub-network Address : 40 . 50 . 92 . 0 / 23

Last (16th) Sub-network Address : 40 . 50 . 94 . 0 / 23

Example 16 :-

Classless network (Network Address : 90 . 152 . 0 . 0 / 14), with 5-bit subnetting.

Network Address : 90 . 152 . 0 . 0 / 14

10011000 . 0 . 0

Broadcast Address : 90 . 155 . 255 . 255 / 14
(Network Directed)

18 bit Host ID

10011011 . 255 . 255

Before Subnetting :-

First Host IP Address : 90 . 152 . 0 . 1 / 14

Last Host IP Address : 90 . 155 . 255 . 254 / 14

Network Size = $(2^{18} - 2)$ hosts



After 5-bit Subnetting :-

Sub-network Address : 90.100110 ^{14 bit NetID} ^{5 bit SubnetID} 00000.000000000 / 19

First Host IP Address : 90.100110 00000.000000001 / 19

Last Host IP Address : 90.100110 11111.111111110 / 19

Broadcast Address : 90.100110 11111.111111111 / 19
(Sub-network Directed)

1st Sub-network Address

: 90.10011000.00000000.0 / 19
 : 90.152.0 . 0 / 19

First Host IP Address

: 90.10011000.00000000.1 / 19
 : 90.152.0 . 1 / 19

Last Host IP Address

: 90.10011000.00011111.254 / 19
 : 90.152.31 . 254 / 19

1st Subnet Broadcast Address

: 90.10011000.00011111.255 / 19

(Sub-network Directed)

: 90.152.31 . 255 / 19

1st subnet $\Rightarrow$ * 90.152.0.0 } 2^8

152.0.255

152.1.0 } 2^8

152.1.255

152.2.0

152.30.255

152.31.0 } 2^8

* 90.152.31.255 } 2^8

$$2^5 \times 2^8 = 2^{13}$$

2nd Sub-network Address

: 90.10011000.00100000.0/19
 : 90.152.32.0/19

First Host IP Address

: 90.10011000.00100000.1/19
 : 90.152.32.1/19

Last Host IP Address

: 90.10011000.00111111.254/19
 : 90.152.63.254/19

2nd Subnet Broadcast Address

: 90.10011000.00111111.255/19

(Sub-network Directed)

: 90.152.63.255/19

Last Sub-network Address

: 90.100110 . 00000.0 / 19
 : 90.155.224.0 / 19

First Host IP Address

: 90.100110 . 00000.1 / 19
 : 90.155.224.1 / 19

Last Host IP Address

: 90.100110 . 11111.254 / 19
 : 90.155.255.254 / 19

Last Subnet Broadcast Address

: 90.100110 . 11111.255 / 19

(Sub-network Directed)

: 90.155.255.255 / 19

After 5-bit Subnetting :-

First Sub-network Address	:	90.152.0.0 / 19	152 → 8
Second Sub-network Address	:	90.152.32.0 / 19	153 → 8
Third Sub-network Address	:	90.152.64.0 / 19	154 → 8
Fourth Sub-network Address	:	90.152.96.0 / 19	155 → 8
5 th 6 th 7 th 8 th 9 th		152.128 152.160 152.192 152.224 90.153.0.0 / 19	
31 th Sub-network Address	:	90.155.192.0 / 19	
Last (32 nd) Sub-network Address	:	90.155.224.0 / 19	



Topic : Subnetting



Suppose,

Net ID field size is 27 bits

Host ID field size is 5 bits

* Fixed
length
subnetting
(prefixing)

Host ID (5 bits)

0	0	0	0	0
0	0	0	0	1
0	0	0	1	0
0	0	0	1	1
0	0	1	0	0
0	0	1	0	1
0	0	1	1	0
0	0	1	1	1
0	1	0	0	0
0	1	0	0	1
0	1	0	1	0
0	1	0	1	1
0	1	1	0	0
0	1	1	0	1
0	1	1	1	0
0	1	1	1	1

1	0	0	0	0
1	0	0	0	1
1	0	0	1	0
1	0	0	1	1
1	0	1	0	0
1	0	1	0	1
1	0	1	1	0
1	0	1	1	1
1	1	0	0	0
1	1	0	0	1
1	1	0	1	0
1	1	0	1	1
1	1	1	0	0
1	1	1	0	1
1	1	1	1	0
1	1	1	1	1



Topic : VLSM



- VLSM : Variable Length Subnet Mask
- Variable Length Network prefixes
[Unlike fixed length prefixing]
- Allow variable size sub-networks (subnets)
[efficient utilization of IP address space]



Topic : VLSM

Suppose,

Net ID field size is 27 bits

Host ID field size is 5 bits

Host ID (5 bits)

0	0	0	0	0
0	0	0	0	1
0	0	0	1	0
0	0	0	1	1
0	0	1	0	0
0	0	1	0	1
0	0	1	1	0
0	0	1	1	1
0	1	0	0	0
0	1	0	0	1
0	1	0	1	0
0	1	0	1	1
0	1	1	0	0
0	1	1	0	1
0	1	1	1	0
0	1	1	1	1

1	0	0	0	0
1	0	0	0	1
1	0	0	1	0
1	0	0	1	1
1	0	1	0	0
1	0	1	0	1
1	0	1	1	0
1	0	1	1	1
1	1	0	0	0
1	1	0	0	1
1	1	0	1	0
1	1	0	1	1
1	1	1	0	0
1	1	1	0	1
1	1	1	1	0
1	1	1	1	1





Topic : VLSM

Suppose,

Net ID field size is 27 bits

Host ID field size is 5 bits

Host ID (5 bits)

0 0 0 0 0

0 0 0 0 1

0 0 0 1 0

0 0 0 1 1

0 0 1 0 0

0 0 1 0 1

0 0 1 1 0

0 0 1 1 1

0 1 0 0 0

0 1 0 0 1

0 1 0 1 0

0 1 0 1 1

0 1 1 0 0

0 1 1 0 1

0 1 1 1 0

0 1 1 1 1

1 0 0 0 0

1 0 0 0 1

1 0 0 1 0

1 0 0 1 1

1 0 1 0 0

1 0 1 0 1

1 0 1 1 0

1 0 1 1 1

1 1 0 0 0

1 1 0 0 1

1 1 0 1 0

1 1 0 1 1

1 1 1 0 0

1 1 1 0 1

1 1 1 1 0

1 1 1 1 1





Topic : VLSM

Suppose,

Net ID field size is 27 bits

Host ID field size is 5 bits

Host ID (5 bits)

0	0	0	0	0
0	0	0	0	1
0	0	0	1	0
0	0	0	1	1
0	0	1	0	0
0	0	1	0	1
0	0	1	1	0
0	0	1	1	1
0	1	0	0	0
0	1	0	0	1
0	1	0	1	0
0	1	0	1	1
0	1	1	0	0
0	1	1	0	1
0	1	1	1	0
0	1	1	1	1

1	0	0	0	0
1	0	0	0	1
1	0	0	1	0
1	0	0	1	1
1	0	1	0	0
1	0	1	0	1
1	0	1	1	0
1	0	1	1	1
1	1	0	0	0
1	1	0	0	1
1	1	0	1	0
1	1	0	1	1
1	1	1	0	0
1	1	1	0	1
1	1	1	1	0
1	1	1	1	1



Example 17 :-

Classless network (Network Address : $120.150.0.0/16$)

[Wishes to divide the network into 3 subnets, one with 50% hosts and remaining two with 25% hosts.]

Network Address : $120.150.0.0/16$

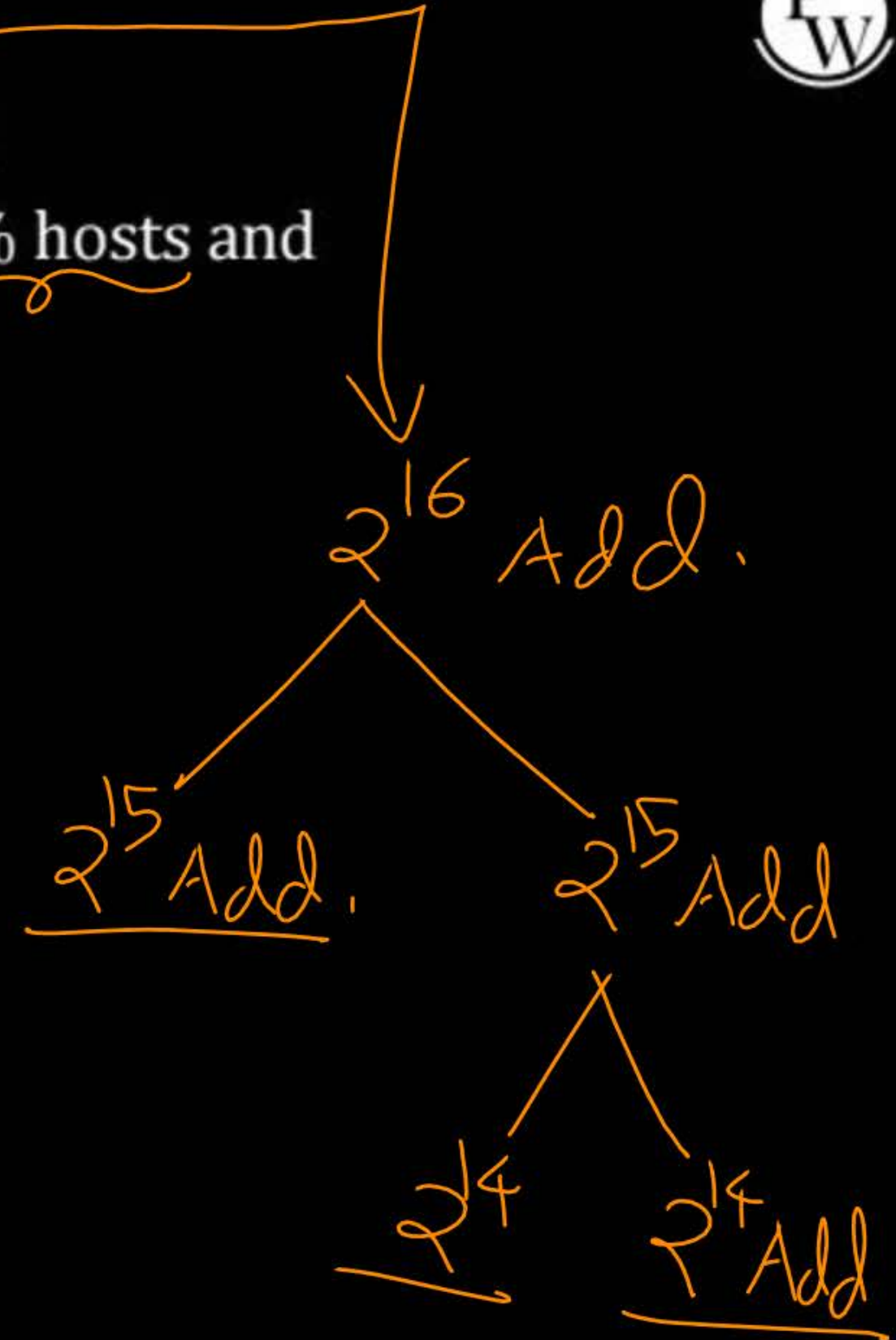
Broadcast Address : $120.150.255.255/16$
(Network Directed)

Before Subnetting :-

First Host IP Address : $120.150.0.1/16$

Last Host IP Address : $120.150.255.254/16$

Network Size = $(2^{16} - 2)$ hosts



Classless network (Network Address : 120.150.0.0 / 16)

120.150.00000000.00000000 / 16
 prefix 16 bit 16 bit Host ID

Subnet-1 Address : 120.150.00000000.00000000 / 17
 [for 50% hosts]
 17 bit prefix 15 bit Host ID

Subnet-2 Address : 120.150.10000000.00000000 / 18
 [for 25% hosts]
 18 bit prefix 14 bit Host ID

Subnet-3 Address : 120.150.11000000.00000000 / 18
 [for 25% hosts]
 18 bit prefix 14 bit Host ID

Classless network (Network Address : 120 . 150 . 0 . 0 / 16)

120 . 150 . 0 0 0 0 0 0 0 0 . 0 0 0 0 0 0 0 0 / 16
 16 bit prefix 16 bit Host ID

Subnet-1 Address : 120 . 150 . 1 0 0 0 0 0 0 0 . 0 0 0 0 0 0 0 0 / 17
 [for 50% hosts]
 17 bit prefix 15 bit Host ID

Subnet-2 Address : 120 . 150 . 0 0 0 0 0 0 0 0 . 0 0 0 0 0 0 0 0 / 18
 [for 25% hosts]
 18 bit prefix 14 bit Host ID

Subnet-3 Address : 120 . 150 . 0 1 0 0 0 0 0 0 . 0 0 0 0 0 0 0 0 / 18
 [for 25% hosts]
 18 bit prefix 14 bit Host ID

Subnet-1 Address : 120 . 150 . 0 . 0 / 17
Subnet-2 Address : 120 . 150 . 128 . 0 / 18
Subnet-3 Address : 120 . 150 . 192 . 0 / 18

Subnet-1 Address : 120 . 150 . 128 . 0 / 17
Subnet-2 Address : 120 . 150 . 0 . 0 / 18
Subnet-3 Address : 120 . 150 . 64 . 0 / 18

Subnet-1 Address : 120 . 150 . 0 . 0 / 17
Subnet-2 Address : 120 . 150 . 192 . 0 / 18
Subnet-3 Address : 120 . 150 . 128 . 0 / 18

Subnet-1 Address : 120 . 150 . 128 . 0 / 17
Subnet-2 Address : 120 . 150 . 64 . 0 / 18
Subnet-3 Address : 120 . 150 . 0 . 0 / 18

#Q. A company has a class C network address of “204 . 204 . 204 . 0” . It wishes to have three subnets, one with 100 hosts and two with 50 hosts each. Which one of the following options represents a feasible set of subnet address/subnet mask pairs?

- (A) 204 . 204 . 204 . 128 / 255 . 255 . 255 . 192
 204 . 204 . 204 . 0 / 255 . 255 . 255 . 128
 204 . 204 . 204 . 64 / 255 . 255 . 255 . 128
- (B) 204 . 204 . 204 . 0 / 255 . 255 . 255 . 192
 204 . 204 . 204 . 192 / 255 . 255 . 255 . 128
 204 . 204 . 204 . 64 / 255 . 255 . 255 . 128
- (C) 204 . 204 . 204 . 128 / 255 . 255 . 255 . 128
 204 . 204 . 204 . 192 / 255 . 255 . 255 . 192
 204 . 204 . 204 . 224 / 255 . 255 . 255 . 192
- (D) 204 . 204 . 204 . 128 / 255 . 255 . 255 . 128
 204 . 204 . 204 . 64 / 255 . 255 . 255 . 192
 204 . 204 . 204 . 0 / 255 . 255 . 255 . 192

[GATE-IT-2005, 2 Marks]

IIT-B, H.W.

MCQ



2 mins Summary



Topic

VLSM



THANK - YOU

